

Instructions: 1) Preferably write the answers following the order of questions. 2) You MUST SHOW ALL the relevant steps of your calculation/derivation. 3) Clearly MARK the FINAL answer. 4) Use standard mathematical notations/symbols only. 5) All graphs should be suitably labelled and axes marked. 6) Marks will be deducted for irrelevant calculations/derivations. 7) No marks will be given for a partially correct answer.

Q1. What is the difference between incoherent and coherent feedforward network motifs? Explain briefly along with their diagrams. [2]

Q2. Two proteins (x and y) control each other through mutual repression. The following system of ODEs is used to model this system. It has only one steady state. Analyze the stability of this steady state and identify the type of phase portrait expected for this steady state. [4]

$$\frac{dx}{dt} = \frac{y}{1+y} - rx, \quad \frac{dy}{dt} = \frac{x}{1+x} - y, \text{ for } x \geq 0, y \geq 0, r > 3$$

Q3. Solve the ODE $\frac{dy}{dt} = -2y + 3$ using the Euler method with step size $\Delta t = 0.2$ in the interval $0 \leq t \leq 0.6$ and report the value of y at $t = 0.6$. Given, at $t = 0$, $y = 1$. [4]

Q4. A protein autoregulates its own expression. We use the ODE $\frac{dP}{dt} = \frac{\alpha}{1 + \left(\frac{P}{K}\right)^n} - \beta P$ to capture the dynamics

of this system. Here, $P \geq 0$, $\alpha > 0$, $\beta > 0$ and $K > 0$. Will any steady state of this system be stable? [5]

Q5. We have a regulatory circuit involving two genes – X and Y . X is produced constitutively at a constant rate. Y induces degradation of X , such that X degrades only in the presence of Y . X induces expression of Y . There is no production of Y in the absence of X . Create a simple but logical ODE-based model for this system. Will this system have a stable steady state? You have to justify your answer. [5]

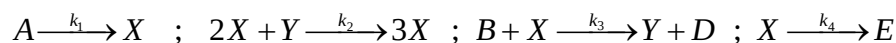
Q6. The following system of ODEs is a simplified version of the FitzHugh–Nagumo model that is used to model neuronal activity. Show that this system has bifurcation, with r being the bifurcation parameter. [5]

$$\frac{dx}{dt} = x - x^3 - y + I, \quad \frac{dy}{dt} = 0.01x + r - y, \text{ for } x \geq 0, y \geq 0, r \geq 1, I \geq 2$$

Q7. The following system of ODEs represents a biomolecular system. It has a steady state $x = y = 1$. Find the relation between p and q that assures periodic oscillation around the steady state in the system.

$$\frac{dx}{dt} = 1 - xy^p, \quad \frac{dy}{dt} = qy(xy^{p-1} - 1), \text{ for } x \geq 0, y \geq 0, q > 0, p > 1$$
 [5]

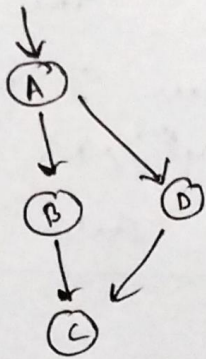
Q8. The following reaction scheme represents an autocatalytic chemical system.



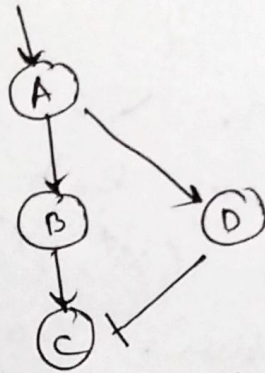
Create a dynamical model for this system following the Law of Mass Action. Assume that A and B are in large excess. That means A and B are constants (i.e., their concentrations do not change with time). [5]

Q1. In feedforward motif input signals takes two paths to reach a common downstream node. In coherent feedforward both the paths have same logic. But in an incoherent feedforward then two paths have opposite logic.

Example



Coherent feedforward



Incoherent feedforward.

Q2.

$$\frac{dx}{dt} = \frac{y}{1+y} - \gamma x$$

$$\frac{dy}{dt} = \frac{x}{1+x} - \gamma y$$

$$\left. \begin{array}{l} x > 0, y > 0 \\ \gamma > 3 \end{array} \right\}$$

The Jacobian,

$$J = \begin{vmatrix} -\gamma & \frac{1}{(1+y)^2} \\ \frac{1}{(1+x)^2} & -\gamma \end{vmatrix}$$

PTS $\rightarrow$

(1)

Q2. Continued...

$$\text{Tr } J = -(\gamma + 1) < 0 \quad \text{as } \gamma > 3$$

$$\det J = \gamma - \underbrace{\frac{1}{(1+n)^2} \cdot \frac{1}{(1+y)^2}}_{\substack{\text{positive and} \\ \text{less-equal to 1,} \\ n, y > 0}} > 0, \quad \text{as } \gamma > 3$$

As $\text{Tr } J < 0$ & $\det J > 0$, the real part of both the eigenvalue of Jacobian J at (x^*, y^*) , where $x^*, y^* > 0$, will be negative. So the steady state will be stable. (Ans) — (1)

$$\begin{aligned} (\text{Tr } J)^2 - 4 \det J &= (\gamma + 1)^2 - 4 \left[\gamma - \frac{1}{(1+n)^2(1+y)^2} \right] \\ &= \gamma^2 + 2\gamma + 1 - 4\gamma + \frac{4}{(1+n)^2(1+y)^2} \\ &= \underbrace{(\gamma - 1)^2}_{> 0 \text{ as } \gamma > 3} + \underbrace{\frac{4}{(1+n)^2(1+y)^2}}_{> 0} \end{aligned}$$

$$\therefore (\text{Tr } J)^2 - 4 \det J > 0$$

$$\Rightarrow (\text{Tr } J)^2 > 4 \det J$$

$\therefore$ Both the eigenvalues are real.

$\therefore$ The phase portrait around the steady state will be Node sink. (Ans) — (2)

Q3.

$$\frac{dy}{dt} = -2y + 3; \quad \Delta t = 0.2$$

$$t=0 \Rightarrow y=1$$

t	y(t)	dy/dt	y(t) + $\frac{dy}{dt} \times \Delta t$
0	1	1	1.2
0.2	1.2	0.6	1.32
0.4	1.32	0.36	1.392
0.6	1.392		

$$\text{At } t = 0.6, y = 1.392 \text{ Am}$$

Q4.

$$\frac{dp}{dt} = \frac{\alpha}{1 + \left(\frac{p}{k}\right)^n} - \beta p = f(p)$$

$$p > 0, \alpha > 0, \beta > 0, k > 0$$

To check stability of a steady state
evaluate $\frac{d}{dp} f(p) = f'(p)$ at a
steady state (p^*)

$$\frac{d}{dp} f(p) = f'(p) = \frac{-\alpha n \left(\frac{p}{k}\right)^{n-1}}{p \left[1 + \left(\frac{p}{k}\right)^n\right]^2} - \beta$$

$$p^* = 0 \text{ then } f'(p^*) = -\beta < 0$$

PTO $\rightarrow$

Q4. Continue...

If $p^* > 0$, ~~the system is stable~~

the $f'(p^*) < 0$

Important condition: $n > 0$

∴ The steady state is stable. Ans

Q5. Check the last page for the correct solution of Q5

$\frac{dx}{dt} = \dots$ This is

$\frac{dy}{dt} = \dots - k_4 y$

the simplest but logically correct model

You can create other models for the same; but that must be logically correct.

Jacobian matrix of the system,

$$J = \begin{vmatrix} -k_2 y & -k_2 x \\ k_3 y & k_3 x - k_4 \end{vmatrix}$$

(x^*, y^*) , where $x^*, y^* > 0$.

Let $k_1, k_2, k_3, k_4 > 0$.

$$J_{(x^*, y^*)} = \begin{vmatrix} -k_2 y^* & -k_2 x^* \\ k_3 y^* & k_3 x^* - k_4 \end{vmatrix}$$

Check the last page for the correct solution of Q5

Q5. continued

∴ the ODEs for $\frac{dx}{dt} = 0$ & $\frac{dy}{dt} = 0$,

We get the steady state

$$x^* = \frac{k_4}{k_3}, \quad y^* = \frac{k_1 k_3}{k_2 k_4}$$

$$\therefore J(x^*, y^*) = \begin{vmatrix} -\frac{k_1 k_3}{k_4} & -\frac{k_2 k_4}{k_3} \\ \frac{k_1 k_3^2}{k_2 k_4} & 0 \end{vmatrix}$$

$$\therefore \text{Tr } J = -\frac{k_1 k_3}{k_4} < 0 \quad \text{as } k_1, k_2, k_3, k_4 > 0$$

$$\det J = k_1 k_3 > 0$$

As $\text{Tr } J < 0$ & $\det J > 0$, this steady state will be stable. [Real part of both eigenvalues of J will be negative]. Ans

This steady state could be either node sink or spiral sink. We can check that but

that will depend upon values of k_1, k_2, k_3, k_4 .

OC.

$$\frac{dx}{dt} = x - x^3 - y + I \quad \text{--- (1)}$$

$$\frac{dy}{dt} = 0.01x + r - y \quad \text{--- (2)}$$

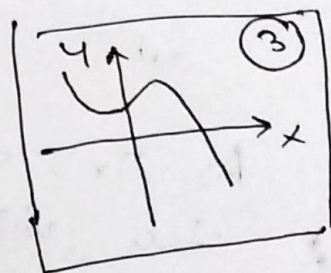
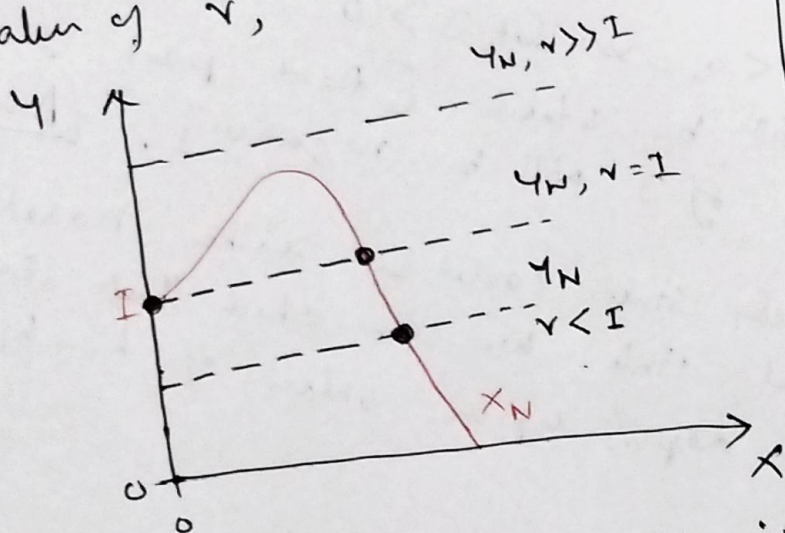
~~No steady state~~,
~~From (1) & (2)~~ Nullcline of x , $\frac{dx}{dt} = 0 \Rightarrow$

$$y = x - x^3 + I \quad \text{--- (3)}$$

Nullcline of y ,

$$\frac{dy}{dt} = 0 \Rightarrow y = 0.01x + r \quad \text{--- (4)}$$

r affects only the y -nullcline.
 Plot the nullcline plot for 3 different values of r ,



From the nullcline plot, it is evident that number of steady state changes with change in r .

$\xrightarrow{PTO}$

(c)

Q6. continued...

Therefore, this system has bifurcation
and r is the bifurcation parameter.

Ans

Q7. $\frac{dx}{dt} = 1 - xy^p$

$$\frac{dy}{dt} = qy(xy^{p-1} - 1)$$

$$= qxy^p - qy$$

The steady state is $x^* = y^* = 1$

At this steady state, the Jacobian is,

$$J = \begin{vmatrix} -y^p & -pxy^{p-1} \\ qy^p & pqxy^{p-1} - q \end{vmatrix}$$

$$= \begin{vmatrix} -1 & -p \\ q & pq - q \end{vmatrix}$$

To have periodic oscillation around this
steady state, we need $\det J > 0$ and

$$\text{Tr } J = 0$$

$p > 0 \rightarrow$

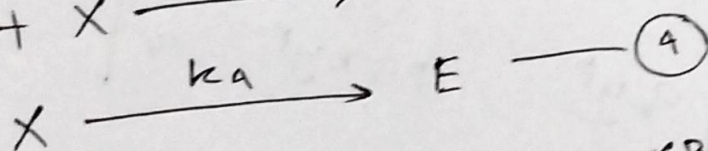
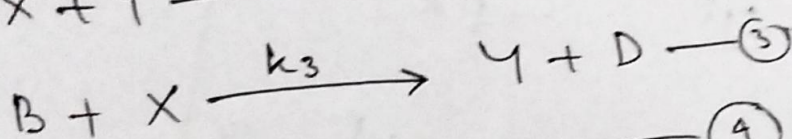
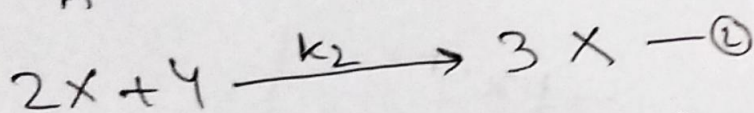
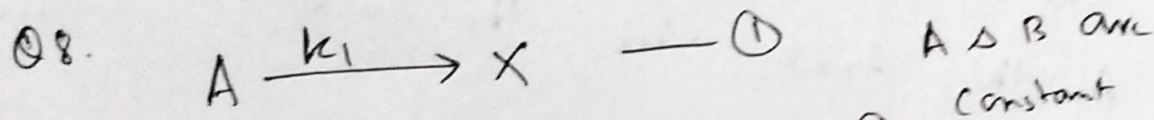
$\det J = q_v > 0$; This condition is already given in question.

$$T_v J = p q_v - q_v - 1$$

As $T_v J = 0$,

$$q_v = \frac{1}{(p-1)}$$

$\therefore$ When $q_v = \frac{1}{(p-1)}$, there will be periodic oscillation ~~about~~ around the steady state (1, 1). Ans



$$\frac{dX}{dt} = \frac{k_1 A}{\text{From (1)}} + \frac{k_2 X^2 Y}{\text{From (2)}} - \frac{k_3 X B}{\text{From (3)}} - \frac{k_4 X}{\text{From (4)}}$$

$$\frac{dY}{dt} = \frac{k_3 X B}{\text{From (3)}} - \frac{k_2 X^2 Y}{\text{From (2)}} ;$$

$$\frac{dD}{dt} = k_3 X B ; \quad \frac{dE}{dt} = k_4 X$$

Q5

$$\left\{ \begin{aligned} \frac{dx}{dt} &= k_1 - k_2 xy \\ \frac{dy}{dt} &= k_3 x - k_4 y \end{aligned} \right.$$

At steady state $\frac{dx}{dt} = \frac{dy}{dt} = 0$.

$$\therefore x^* = \sqrt{\frac{k_1 k_4}{k_2 k_3}}, \quad y^* = \left(\frac{k_3}{k_4}\right) \sqrt{\frac{k_1 k_4}{k_2 k_3}}$$

$$\therefore x^*, y^* > 0$$

The Jacobian at steady state

$$J = \begin{vmatrix} -k_2 y^* & -k_2 x^* \\ k_3 & -k_4 \end{vmatrix}$$

$$\det(J) = k_2 k_4 y^* + k_2 k_3 x^* > 0$$

$$\text{Tr}(J) = -(k_2 y^* + k_4) < 0$$

So, the real part of both the eigenvalues of J must be negative.

$\therefore$ The stability of the steady state must be either nodal sink (stable) or spiral sink (stable).

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